ARAVIND NAIDU YALLA

AI Software Engineer – Production AI Systems, Python & End-to-End Delivery

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SUMMARY

AI Software Engineer with 1 year of hands-on experience shipping production software end-to-end – from scoping and architecture through deployment and maintenance. Built and deployed Generative AI applications (RAG, LLM integration, LangChain/LangGraph) and backend systems (Python, FastAPI, PostgreSQL) that replaced manual workflows and delivered measurable results, including a 50% production performance improvement. Comfortable working with ambiguous requirements, translating business needs into shipped features, and owning a workstream from design decisions through to what a team actually relies on daily. Strong CS fundamentals, systematic debugging, and a bias toward outcomes: focused on whether the thing built actually works, not just whether it was built.

SKILLS

Agentic AI & LLM: LLMs, LangChain, LangGraph, RAG Pipelines, Tool/Function Calling, Prompt Engineering, LLM Output Evaluation, Hugging Face Transformers, Ollama / Local Models (Llama)

Vector & Databases: Qdrant (Vector DB), Embeddings, Semantic Search, PostgreSQL, SQL Optimization

Languages: Python (primary), JavaScript, HTML, CSS

Backend & APIs: FastAPI, Flask, REST API Design, Pydantic, SQLAlchemy, OOP, Design Patterns

Engineering Practices: Git/GitHub, Debugging & Troubleshooting, Code Reviews

Deployment: Docker (basics), IIS Server, NSSM Service Automation, GitHub Copilot

EXPERIENCE

AI Developer – People Media Factory, Hyderabad, India *May 2026 – Present (Full-Time) · Oct 2025 – Apr 2026 (Intern)*

- Owned features end-to-end from scoping through deployment, translating ambiguous business requirements into shipped, production-grade software
- Built and shipped LLM-powered and agentic features into production systems, integrating RAG pipelines and prompt-engineered workflows using LangChain and LangGraph
- Designed OOP-based Python services and REST APIs, collaborating directly with cross-functional stakeholders to turn business needs into working solutions
- Diagnosed and resolved a production performance defect through systematic debugging, optimizing PostgreSQL queries for 50% faster performance
- Integrated Hugging Face Transformers and locally deployed models (Ollama/Llama), and evaluated LLM/RAG output quality before production rollout
- Deployed and maintained services on IIS Server with NSSM automation, taking ownership of solutions a team relies on daily
- Managed codebase via Git branching workflows; participated in code reviews and maintained technical documentation

KEY PROJECTS

AI Chatbot with RAG – Document Intelligence *Python, FastAPI, LangChain, LangGraph, Qdrant, Hugging Face*

- Built a RAG application end-to-end: document/OCR pipeline, embeddings, Qdrant vector store, LangGraph-orchestrated retrieval, and a FastAPI backend for document Q&A
- Added function-calling logic so the LLM selects tools/retrieval steps dynamically instead of following a fixed pipeline
- Ran evaluation passes on generated responses to catch hallucinations and improve retrieval relevance

VFX Production Tracking System *Python, FastAPI, PostgreSQL, Git, IIS, NSSM*

- Built a production-grade tracking system (REST APIs, RBAC, analytics dashboards) across the full SDLC, from design through deployment
- Diagnosed and resolved a production performance defect through systematic debugging and SQL optimization, improving response times by 50%

EDUCATION & CERTIFICATIONS

B.Tech in Computer Science (Data Science) – Kakinada Institute of Engineering and Technology 2025 · CGPA: 7.5

Certifications: Python for Data Science (Infosys Springboard) · Salesforce Developer (SmartInternz)